

Academic Planning OS

Optimal Data Acquisition, Scraping & Verification Specification

Engineering handoff for collecting university, major, course, transfer, exam-credit, residency, and program-policy data

Primary goal: Build a source-backed academic policy dataset that is complete enough for GPT-5.6 to answer and simulate student plans without inventing equivalencies, prerequisites, or admissions requirements.

1. Scope and Operating Principles

- The scraper must treat each institution as a network of related sources, not as a single webpage.
- Every extracted fact must retain its source URL, page title, effective catalog year or term, retrieval timestamp, and confidence status.
- Official university sources take priority over third-party sources. Student forums may help locate an official page but must not become authoritative policy data.
- Rules must be represented as structured data where possible and preserved as source text when interpretation is needed.
- Conflicts, ambiguity, stale pages, and missing information must be surfaced rather than silently resolved.
- The system must distinguish admission eligibility, degree applicability, transferability, and course enrollment prerequisites. These are separate questions.

Critical distinction: A course can transfer for credit but fail to satisfy a major prerequisite. It can satisfy a general-education category but not a specific course. It can count toward admission eligibility but not toward graduation. Store these outcomes separately.

2. Source Hierarchy: What to Trust First

Priority	Source type	Use for	Handling rule
1	Official catalog / academic bulletin	Courses, credits, prerequisites, degree requirements, catalog year	Primary structured source; preserve historical catalog versions.
1	Official admissions and transfer pages	Minimum credits, deadlines, applicant type, residency-related pathways, major admission rules	Capture page-specific effective term and audience.

1	Official department / major website	Major prerequisites, application cycles, capacity constraints, sequencing, recommended preparation	Compare against catalog; flag conflicts.
1	Official transfer-equivalency or articulation database	Direct equivalents, transfer categories, sending-institution rules	Prefer API or export where available; retain date range and sending institution.
1	Official AP / IB / CLEP credit tables	Score thresholds, course awards, gen-ed treatment, duplicate-credit restrictions	Store by exam, score, school/college, and catalog year.
1	Official general-education pages	Writing, diversity, arts/humanities, quantitative, science, language categories	Capture category definitions and overlap/double-counting rules.
1	Official registrar / residency / tuition pages	Residency classification, credit limits, repeat rules, graduation residency	Do not infer legal residency; store conditions and direct users to official review.
2	Statewide articulation systems and public transfer agreements	Common course numbering, guaranteed pathways, transfer degrees	Use when officially sanctioned by participating institutions.
2	Official PDFs, handbooks, advising sheets, curriculum maps	Major checklists, sample plans, special notes	Extract text and tables; preserve document URL and revision date.
3	Accredited external course databases or archived official pages	Gap filling and historical verification	Mark as secondary and never override current official policy.
4	Reddit, forums, blogs, advisor anecdotes	Discovery only	Never present as policy or use for automatic decisions.

3. Institution Crawl Strategy

For each school, begin with a domain map and then crawl by policy family. Do not rely on broad recursive crawling alone.

1. Identify the canonical academic catalog domain and current catalog year.
2. Identify admissions, registrar, transfer-credit, general-education, advising, and departmental subdomains.
3. Collect the institution's sitemap.xml, robots.txt, catalog index, A–Z index, and search endpoint where permitted.
4. Generate a seed URL list from official navigation rather than guessing URLs.

5. Follow links only within allowed official domains and explicitly whitelisted state articulation systems.
6. Classify every page before extraction: course, major, admissions, transfer, exam credit, gen-ed, residency, graduate program, study abroad, or miscellaneous.
7. Record redirects and canonical URLs so duplicate pages do not create conflicting records.
8. Detect PDFs, embedded catalogs, JavaScript-rendered databases, accordions, pagination, and downloadable CSVs.
9. Build a page relationship graph so a major page can point to its prerequisite courses, admission page, sample plan, and degree requirements.

4. Page Discovery Checklist and Search Vocabulary

Search both site navigation and the institution's internal search for these terms and close variants:

- Transfer admission; transfer requirements; transfer applicant; lower-division transfer; upper-division transfer
- Credit transfer; transfer credit; course equivalency; articulation; transferology; prior learning
- Degree requirements; graduation requirements; university requirements; residency credits
- General education; core curriculum; breadth; distribution; areas of inquiry; diversity; writing intensive
- Major requirements; admission to major; capacity-constrained; selective major; pre-major; declaration
- Prerequisite; co-requisite; placement; minimum grade; sequence; permission of instructor
- AP credit; IB credit; CLEP; A-level; Cambridge; dual enrollment; Running Start
- Community college pathway; direct transfer agreement; associate degree; statewide transfer degree
- Catalog; bulletin; curriculum; program sheet; degree map; four-year plan; sample schedule
- Graduate prerequisite; MBA prerequisite; admissions requirements; recommended preparation
- Study abroad equivalency; pre-approval; course petition; international transfer credit
- Residency; in-state tuition; domicile; tuition classification; WUE
- Maximum transfer credits; minimum transfer credits; junior standing; sophomore transfer
- Repeated course; duplicate credit; cross-listed; mutually exclusive; credit by examination

5. Required Data Entities

Institution: ID, legal name, common name, system/campus, state, country, institution type, academic calendar, official domains, catalog years.

Program/Major: Degree type, college/school, campus, admission type, capacity status, application terms, minimum GPA, prerequisite groups, degree requirements, residency requirements.

Course: Subject, number, title, description, credits, variable-credit range, level, prerequisites, co-requisites, restrictions, repeatability, equivalent/cross-listed courses, terms offered.

Requirement: Requirement category, rule type, minimum credits/courses, allowed course list, grade floor, overlap rules, residency rule, expiration/effective dates.

Transfer Rule: Sending institution, source course, receiving institution, target equivalent or category, credit amount, effective date range, minimum grade, notes, direct-equivalent confidence.

Exam Credit Rule: Exam type, exam name, score range, awarded course or category, credits, placement effect, duplicate-credit rule, program-specific exception.

Admissions Rule: Applicant type, minimum/preferred credits, deadlines, GPA floor, prerequisite completion timing, residency/pathway conditions, required materials.

General Education Category: Name, description, credit/course requirement, approved courses, overlap rules, upper-division requirement, institution/college applicability.

Academic Calendar/Catalog Version: Academic year, term system, effective date, archived URL, superseded status.

Evidence Record: Source URL, page title, quoted source text or table row, retrieval timestamp, parser version, checksum, authority tier, confidence, reviewer status.

6. Major and Program Website Crawl

- ☐ Locate the official program overview page.
- ☐ Locate the catalog degree-requirements page for the same program and catalog year.
- ☐ Determine whether the major is open, declaration-based, selective, capacity-constrained, direct-admit, or application-based.
- ☐ Capture every application cycle, deadline, start term, and prerequisite completion deadline.
- ☐ Capture minimum overall GPA, prerequisite GPA, course-grade floors, credit minimums, and residency restrictions.
- ☐ Extract prerequisite groups exactly: all-of, any-of, choose-N, sequence, or conditional alternatives.
- ☐ Capture required supporting courses outside the department, such as calculus, chemistry, writing, or statistics.
- ☐ Capture recommended but not required preparation separately from mandatory requirements.
- ☐ Find sample plans, advising sheets, FAQs, info-session slides, and application rubrics.
- ☐ Capture progression rules after admission: continuation GPA, gateway courses, time limits, and repeat limits.
- ☐ Capture degree totals, upper-division totals, major residency, capstone, internship, and experiential requirements.
- ☐ Check whether requirements differ by campus, degree type, concentration, or student entry year.
- ☐ Compare department wording against the official catalog and create a conflict record when they differ.

7. Course Catalog Extraction

- Extract subject code, course number, title, credits, description, prerequisites, co-requisites, restrictions, and repeatability.
- Preserve logical prerequisite structure. Example: “MATH 124 and (CSE 121 or CSE 122)” must not become a flat list.
- Detect minimum grades, concurrent-enrollment permissions, placement alternatives, and “or equivalent” language.
- Record enrollment restrictions such as major-only, class standing, campus, honors, instructor permission, or cohort membership.
- Record cross-listed and mutually exclusive courses to prevent duplicate-credit recommendations.
- Record variable credit, lecture/lab pairings, linked sections, and courses that require both lecture and laboratory.
- Capture term availability only when officially published and store it as probabilistic or historical rather than guaranteed.
- Preserve course version by catalog year because course numbers, credits, and prerequisites change.
- Link each course to the requirements and programs it can satisfy, but keep the source evidence for every link.

8. Transfer Equivalency and Articulation Extraction

Do not collapse outcomes: Store direct course equivalency, departmental elective, general elective, gen-ed category, and no-credit outcomes as different result types.

- ☐ Sending institution and campus must be explicit. A course number is not globally unique.
- ☐ Capture source course subject, number, title, credits, and date range.
- ☐ Capture receiving course equivalent, credit value, requirement category, or elective designation.
- ☐ Capture minimum transferable grade, pass/fail treatment, lab requirement, sequence requirement, and credit caps.
- ☐ Capture one-to-many and many-to-one equivalencies, such as a two-course sequence equaling one receiving course.
- ☐ Capture conditional mappings, such as “with lab,” “for majors only,” “before Fall 2025,” or “pending syllabus review.”
- ☐ Capture transfer degree guarantees and community-college-specific pathways separately from course equivalencies.
- ☐ Check whether AP/IB credit awarded at the sending institution is re-evaluated independently by the receiving institution.
- ☐ Check maximum lower-division, upper-division, community-college, and total transfer credits.
- ☐ Check whether transfer credit counts toward class standing, admission eligibility, degree credit, and major requirements independently.
- ☐ When no equivalency exists, store “not found,” not “does not transfer.” The latter requires an explicit official denial.

9. General Education and University Requirements

- ☐ Capture all category names and aliases, such as Arts & Humanities, Social Sciences, Natural Sciences, Diversity, Writing, Quantitative, Language, and First-Year Seminar.
- ☐ Capture minimum credits, minimum number of courses, upper-division requirements, lab requirements, and approved-course lists.
- ☐ Capture whether one course may satisfy multiple categories and whether credits can be double-counted.
- ☐ Capture college-specific general education rules; engineering, business, honors, and liberal arts may differ.
- ☐ Capture transfer-degree waivers, completed-core guarantees, and exceptions for students entering with an associate degree.
- ☐ Capture writing-in-the-major, diversity, capstone, and residency requirements that are often omitted from simple degree totals.
- ☐ Preserve the catalog year that governs the user because approved-course lists can change.

10. AP, IB, CLEP, A-Level, and Prior Learning Credit

- ☐ Extract each exam separately by exam name and score band.
- ☐ Capture awarded course, credits, placement, general-education category, and whether the award differs by college or major.
- ☐ Capture duplicate-credit rules when the student later takes the equivalent college course.

- ☐ Capture whether exam credit satisfies admission prerequisites, major prerequisites, graduation requirements, or only placement.
- ☐ Capture score-reporting and deadline requirements where relevant.
- ☐ Capture special cases where a score earns credit for non-majors but not for majors.
- ☐ If the page is silent about using exam credit for a selective-major prerequisite, mark it unresolved and trigger an admissions/department verification recommendation.

11. Admissions, Applicant Classification, and Residency

- ☐ Minimum and preferred number of transferable credits at application and at enrollment.
- ☐ How the institution classifies applicants with AP, dual-enrollment, Running Start, or post-high-school college work.
- ☐ Differences between community-college transfer, four-year transfer, in-state, out-of-state, international, returning, and second-bachelor applicants.
- ☐ Whether certain pathways or guarantees apply only to residents or partner colleges.
- ☐ Maximum credits accepted and whether excess credits are lost, held in reserve, or still used for prerequisites.
- ☐ Major-specific admission versus university admission and whether a student can enter undeclared.
- ☐ Prerequisite completion timing: before application, by a specific term, or before enrollment.
- ☐ Application materials, essays, portfolios, recommendations, tests, and transcript requirements.
- ☐ Residency policy facts must be presented as official conditions, never as a definitive legal determination by the scraper.

12. Graduate, MBA, and Study-Abroad Data

- ☐ Graduate/MBA: required and recommended undergraduate courses, GPA floors, test policy, work-experience expectations, prerequisite recency, and application deadlines.
- ☐ Graduate/MBA: distinguish program admission requirements from preparation that merely improves competitiveness.
- ☐ Study abroad: pre-approved equivalencies, course-level restrictions, minimum grades, credit conversion, syllabus-review requirements, and pre-approval deadlines.
- ☐ Study abroad: capture whether courses can satisfy major, gen-ed, language, residency, or only elective credit.
- ☐ For both: preserve program-specific rules instead of applying institution-wide assumptions.

13. Verification, Conflict Resolution, and Confidence

Every extracted rule should pass the following checks before it is considered usable for automated simulation:

Check	Question	Failure action
Authority	Is the source official and does it govern this exact institution, campus, applicant type, program, and catalog year?	Lower confidence, create review task, and prevent definitive recommendation.
Freshness	Does the page show an effective date, catalog year, or revision date? Is it current or archived?	Lower confidence, create review task, and prevent definitive recommendation.
Specificity	Does the rule apply institution-wide, to one college, one major,	Lower confidence, create review task, and prevent definitive

	one applicant type, or one sending institution?	recommendation.
Completeness	Were footnotes, expandable sections, table headers, PDF notes, and linked exceptions captured?	Lower confidence, create review task, and prevent definitive recommendation.
Consistency	Does the catalog agree with the department page, admissions page, and transfer database?	Lower confidence, create review task, and prevent definitive recommendation.
Logical integrity	Are all-of, any-of, choose-N, sequence, grade, and timing conditions represented correctly?	Lower confidence, create review task, and prevent definitive recommendation.
Evidence	Can the application show the exact supporting source text or table row to the user?	Lower confidence, create review task, and prevent definitive recommendation.
Ambiguity	Would a reasonable advisor still need to interpret the rule? If yes, mark it unresolved.	Lower confidence, create review task, and prevent definitive recommendation.

14. Recommended Confidence Tiers

Verified: Official source, exact scope, current effective period, structured extraction validated, no material conflicts.

High confidence: Official source and exact scope, but effective date or one secondary detail is unclear.

Needs review: Official sources conflict, rule depends on interpretation, or extraction from a complex PDF/table may be incomplete.

Unresolved: No authoritative answer found. The app should state what is unknown and draft a question for admissions, the registrar, or the department.

Deprecated: Previously valid policy retained for historical transcript analysis but not current planning.

15. Technical Scraping Guidelines

- Respect robots.txt, site terms, rate limits, and access controls. Prefer published APIs, downloadable data, and sitemaps.
- Use identifiable, low-rate requests with caching and exponential backoff. Never bypass authentication, CAPTCHAs, or anti-bot controls.
- Hash page content and reprocess only changed pages.
- Store raw HTML/PDF snapshots or legally permissible normalized text for reproducibility and change detection.
- Render JavaScript only when necessary; prefer underlying JSON endpoints when publicly exposed and permitted.
- Parse tables with headers and footnotes intact. A row without its header context is unsafe.
- For PDFs, extract both text and table structure; send low-confidence or image-only documents to manual review.
- Normalize whitespace and identifiers, but preserve original source text for evidence.
- Deduplicate by canonical URL plus content hash, not page title alone.
- Create adapters by catalog platform when possible, such as Acalog, CourseLeaf, Modern Campus, PeopleSoft catalogs, or custom systems.

- Separate discovery, fetching, parsing, normalization, validation, and publishing into distinct stages.
- Log parser version, crawl job, errors, skipped pages, and changes in policy records.

16. Recommended Data Pipeline

10. Discover official domains, catalog versions, sitemaps, databases, PDFs, and seed pages.
11. Fetch content under domain-specific rate and access rules.
12. Classify each page into a policy family.
13. Extract structured fields and preserve evidence spans.
14. Normalize institution, campus, term, course, program, and requirement identifiers.
15. Resolve links between courses, requirements, majors, transfer rules, and exams.
16. Run schema validation and logical checks.
17. Compare overlapping official sources and create conflict records.
18. Assign confidence and route uncertain records to review.
19. Publish versioned records to the application knowledge base.
20. Monitor source changes and re-run only affected dependencies.

17. Required QA Test Cases

- ☐ A course transfers directly but does not satisfy the target major prerequisite.
- ☐ Two source courses together equal one destination course.
- ☐ One source course maps to two destination courses or multiple requirement categories.
- ☐ AP credit grants placement but no graduation credit.
- ☐ AP credit satisfies a university requirement but not a selective-major prerequisite.
- ☐ A major prerequisite requires a minimum grade higher than the university transfer minimum.
- ☐ A prerequisite has nested logic: A and (B or C).
- ☐ A course is cross-listed and must not award duplicate credit.
- ☐ A transfer agreement is valid only for a specific academic-year range.
- ☐ A student attended multiple institutions with duplicate or repeated courses.
- ☐ A community-college pathway waives gen-ed, but a four-year transfer pathway does not.
- ☐ A major page and catalog disagree.
- ☐ A requirement differs by campus, degree type, concentration, or entry year.
- ☐ A PDF footnote changes the meaning of a table row.
- ☐ No equivalency is listed, requiring syllabus review rather than a “no transfer” conclusion.
- ☐ Excess transfer credits count toward prerequisites but not toward the degree credit limit.
- ☐ A study-abroad course needs pre-approval and cannot be assumed to satisfy the major.

18. School-by-School Completion Checklist

Institution setup

- ☐ Official domains identified
- ☐ Current and archived catalog years identified
- ☐ Academic calendar and campus structure recorded
- ☐ robots.txt and access rules reviewed
- ☐ Sitemaps and seed URLs captured

Admissions

- ☐ First-year requirements
- ☐ Transfer requirements
- ☐ Community-college transfer rules
- ☐ Four-year transfer rules
- ☐ In-state/out-of-state distinctions
- ☐ International transfer rules
- ☐ Minimum/preferred credits
- ☐ Deadlines and required materials

Academics

- ☐ Complete course catalog
- ☐ Prerequisite logic
- ☐ General education
- ☐ Graduation requirements
- ☐ Residency credits
- ☐ Repeat/duplicate-credit rules
- ☐ Maximum transfer credits

Majors

- ☐ Program inventory
- ☐ Major admission type
- ☐ Major prerequisites
- ☐ Minimum grades and GPA
- ☐ Application terms/deadlines
- ☐ Degree requirements
- ☐ Sample plans and advising sheets
- ☐ Concentrations and campus variants

Transfer

- ☐ Equivalency database
- ☐ Articulation agreements
- ☐ Statewide pathways
- ☐ Direct equivalents
- ☐ Elective/category mappings
- ☐ Date ranges
- ☐ Minimum grades
- ☐ Sequence and lab rules

Exam credit

- ☐ AP
- ☐ IB
- ☐ CLEP
- ☐ A-Level/Cambridge
- ☐ Dual enrollment treatment
- ☐ Duplicate-credit restrictions
- ☐ Major-specific exceptions

Extended planning

- ☐ Graduate prerequisites
- ☐ MBA prerequisites/recommendations
- ☐ Study-abroad equivalencies and approval rules

Quality

- ☐ All records have evidence URLs
- ☐ Effective dates/catalog years captured
- ☐ Conflicts logged
- ☐ Unresolved rules marked
- ☐ Change monitoring enabled
- ☐ Representative plans manually spot-checked

19. Definition of Done for One Institution

A school should not be marked production-ready until all of the following are true:

- ☐ A user can upload a transcript and receive a source-backed transfer result for representative courses.
- ☐ The system can distinguish transferable credit, direct equivalency, general education, major prerequisite, elective credit, and non-evaluated credit.
- ☐ The system can evaluate at least the school's most common majors and clearly label unsupported programs.
- ☐ AP/IB treatment is represented and uncertainty about selective-major use is surfaced.
- ☐ Transfer applicant classification, credit minimums, and major admission timing are available.
- ☐ At least three manually constructed student scenarios pass review: community-college transfer, four-year transfer, and current-school degree planning.
- ☐ Every recommendation shown to the user can link back to an official source.
- ☐ No missing equivalency is presented as an official rejection.
- ☐ Conflict and uncertainty states generate a direct recommendation to contact the correct office, with a pre-drafted question.

20. Expected Engineering Deliverables

- ☐ Institution source map and seed URL manifest.
- ☐ Scraper adapters by catalog/database platform.
- ☐ Normalized schemas for courses, programs, requirements, admissions, transfer, and exam credit.
- ☐ Raw evidence store and content-change history.
- ☐ Validation and conflict-detection rules.
- ☐ Manual review queue for ambiguous or conflicting records.
- ☐ Automated test fixtures covering the QA cases above.
- ☐ Coverage dashboard showing which schools, majors, catalog years, and policy families are complete.
- ☐ Runbook for adding a new institution and re-crawling an existing one.

Final product behavior: When the dataset cannot support a confident answer, GPT-5.6 should never guess. It should explain the exact uncertainty, cite what was found, identify the proper office, and draft a concise admissions or departmental email containing the student's course, institution, term, and intended program.

Appendix A. Minimum Evidence Object

evidence_id
source_url
canonical_url
page_title
source_type
authority_tier
institution_id
campus_id (optional)
catalog_year / effective_start / effective_end
retrieved_at
content_hash
parser_version
source_text_or_table_row
structured_fact_id
confidence_status
conflict_ids[]
review_status

Appendix B. Questions the Scraper Must Never Answer by Assumption

- “No equivalency is listed, so the course will not transfer.”
- “The sending school gave AP credit, so the receiving school must accept it.”
- “A course counts for general education, so it also satisfies the major prerequisite.”
- “The student has enough credits, so they are eligible for the major.”
- “In-state tuition means the student qualifies for an in-state transfer guarantee.”
- “A sample four-year plan is a binding requirement.”
- “A recommended course is mandatory.”
- “The current major page applies to students under every catalog year.”